Mixing effects in radiative decays of heavy-strange axial-vector mesons within light-cone QCD sum rules

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Abstract

We calculate the radiative decay widths of $D_{s1}(2460) \to D_s\gamma$, $D_{s1}(2536) \to D_s\gamma$, $B_{s1}(5750) \to B_s\gamma$, and $B_{s1}(5830) \to B_s\gamma$ within light-cone QCD sum rules, taking into account the mixing between the $^1P_1$ and $^3P_1$ configurations. In the bottom-strange sector, we consider the experimentally established $B_{s1}(5830)$ together with the theoretically expected lower 1^+ state, denoted $B_{s1}(5750)$. We obtain

$$\Gamma[D_{s1}(2460) \to D_s\gamma] = 45.9^{+10.1}_{-8.5}\ \text{keV}, \qquad \Gamma[D_{s1}(2536) \to D_s\gamma] = 0.55^{+1.73}_{-0.48}\ \text{keV},$$

$$\Gamma[B_{s1}(5750) \to B_s\gamma] = 12.0^{+2.9}_{-2.3}\ \text{keV}, \qquad \Gamma[B_{s1}(5830) \to B_s\gamma] = 8.9^{+2.4}_{-2.0}\ \text{keV}.$$

The $D_{s1}(2460) \to D_s\gamma$ decay width is substantially larger than that of $D_{s1}(2536) \to D_s\gamma$. This difference originates from $^1P_1$–$^3P_1$ mixing: the two basis-current contributions interfere constructively for the lower state and destructively for the higher state. In the bottom-strange sector, the two widths are comparable within their uncertainties, although the higher state still exhibits destructive interference. These results show that the radiative decay pattern of strange heavy axial-vector mesons is sensitive to the mixing of the two spin configurations.

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I. INTRODUCTION

Radiative transitions of heavy-light mesons provide complementary information on their internal structure. In particular, electromagnetic transitions are sensitive to the spin composition of the states and to the coupling of the photon to both the heavy and light quarks. They therefore complement mass spectroscopy and strong-decay measurements in determining the properties of heavy-light mesons.

The axial-vector ($J^P = 1^+$) heavy-light mesons are particularly interesting in this respect. Since charge conjugation is not a good quantum number for heavy-light systems, the physical states are not pure $^1P_1$ or $^3P_1$ configurations [1]. Instead, spin-dependent interactions mix these two basis states. In the heavy-quark limit, the physical states are classified according to the total angular momentum of the light degrees of freedom, $j_\ell = \frac{1}{2}$ and $j_\ell = \frac{3}{2}$, which correspond to definite linear combinations of the $^1P_1$ and $^3P_1$ configurations. For finite heavy-quark masses, deviations from this limit modify the mixing pattern and consequently affect the radiative transition amplitudes. Since both components contribute to the physical amplitudes, constructive or destructive interference may occur, making the radiative decays sensitive to the mixing between the two configurations.

The charm-strange axial-vector mesons $D_{s1}(2460)$ and $D_{s1}(2536)$ provide a useful system for studying this effect. The $D_{s1}(2460)$ is a very narrow state lying below the D^*K threshold, and its radiative decay to $D_s\gamma$ has been observed both in B decays and in continuum production [2, 3]. The $D_{s1}(2536)$, on the other hand, is well established through hadronic decay modes such as D^*K and $D\pi K$ [4, 5], while its radiative decay to $D_s\gamma$ has not yet been observed. Since the total width of the $D_{s1}(2536)$ is below the MeV scale, a suppressed radiative branching fraction is experimentally challenging to measure [5]. Very recently, the LHCb Collaboration performed a dedicated study of muon-tagged $D_{s1}(2460)^+$ and $D_{s1}(2536)^+$ decays and obtained the upper limit $\Gamma_{\rm tot}[D_{s1}(2460)^+] < 1.28$ MeV at 90% confidence level, adding new experimental information on these axial-vector charm-strange states [6].

In the bottom-strange sector, the $B_{s1}(5830)$ state has been observed in B^*K spectroscopy [7, 8], whereas its radiative transition to $B_s\gamma$ remains unknown. A second, lower $J^P = 1^+$ bottom-strange state is also expected theoretically as the bottom partner of the $D_{s1}(2460)$. Lattice-QCD calculations [9], hadronic-molecule studies [10], and coupled-channel analyses [11] place this state below or near the B^*K threshold, with mass estimates in the range 5.75–5.78 GeV. This lower state has not yet been observed experimentally. In the present work, we denote it by $B_{s1}(5750)$ and adopt the lattice-QCD mass prediction of Ref. [9].

Radiative transitions of heavy-light mesons have been studied using relativistic and relativized quark

models [12–15], potential and nonrelativistic quark models [16–19], an unquenched coupled-channel model [20], vector-meson-dominance (VMD) approaches [21, 22], effective-field-theory and hadronic-molecule approaches [10, 23, 24], and light-cone QCD sum rules [25, 26].

More recently, phenomenological and potential-model analyses [27, 28] have focused on the role of mixing in the radiative decays of the axial-vector states. Within light-cone QCD sum rules, the $D_{s1}(2460) \to D_s\gamma$ transition is predicted to be one of the dominant radiative decay modes of the $D_{s1}(2460)$ [25]. Bondar and Milstein [27, 28] argued that the large difference between the $D_{s1}(2460) \to D_s\gamma$ and $D_{s1}(2536) \to D_s\gamma$ decay widths originates from interference between the $^1P_1$ and $^3P_1$ components of the mixed physical states. The analysis was also extended to the corresponding bottom-strange system, including both the observed $B_{s1}(5830)$ and the predicted lower 1^+ state.

An important question is whether this interference mechanism also emerges within the light-cone QCD sum-rule framework. In a quark-model description, the cancellation can be traced directly to the components of the meson wave function. In light-cone QCD sum rules, the corresponding effect must instead arise from the correlation functions constructed with independent interpolating currents for the $^1P_1$ and $^3P_1$ configurations. It is therefore interesting to examine whether the sum-rule calculation reproduces the same interference pattern.

In this work, we calculate the radiative decays $D_{s1}(2460) \to D_s\gamma$, $D_{s1}(2536) \to D_s\gamma$, $B_{s1}(5750) \to B_s\gamma$, and $B_{s1}(5830) \to B_s\gamma$ within the light-cone QCD sum-rule framework. We retain the strange-quark-mass contributions and employ photon distribution amplitudes up to twist four. The physical axial-vector states are constructed from two independent interpolating currents through the $^1P_1$–$^3P_1$ mixing scheme. This allows us to study directly how the two basis-current contributions interfere in the physical radiative amplitudes. In the bottom-strange sector, the lower $B_{s1}(5750)$ state is included as a theoretical benchmark, while the result for the experimentally established $B_{s1}(5830)$ constitutes a prediction for its radiative decay.

The paper is organized as follows. In Sec. II, we introduce the interpolating currents and derive the light-cone QCD sum rules for the transition form factors. In Sec. III, we present the numerical analysis, compare our results with previous theoretical studies, and discuss the current experimental status. Our conclusions are summarized in Sec. IV.

II. THEORETICAL FRAMEWORK

We consider the radiative transitions

$$A_i(p',\eta) \to P(p)\gamma(q,\varepsilon), \qquad p' = p + q, \qquad i = 1, 2, \tag{1}$$

where A_i is a physical heavy axial-vector meson and P is the corresponding pseudoscalar meson. Here, η and ε denote the polarization vectors of the axial-vector meson and the photon, respectively. The heavy quark is denoted by $Q = c, b$. Since all channels considered here are strange, the light-quark field is denoted by s throughout, while q is reserved exclusively for the photon momentum. Axial-vector and pseudoscalar mesons masses are denoted by m_{A_i} and m_P, respectively.
The transition is described by the following vacuum correlation function:

$$\Pi^{(i)}_{\mu\nu}(p',q) = i^2 \int d^4x \int d^4y\, e^{ip\cdot x + iq\cdot y} \langle 0|T\left\{J^{(i)}_\mu(x) J^{\rm em}_\nu(y) J^\dagger_5(0)\right\}|0\rangle, \tag{2}$$

where

$$J^{\rm em}_\nu = e_s\, \bar{s}\gamma_\nu s + e_Q\, \bar{Q}\gamma_\nu Q \tag{3}$$

is the electromagnetic current, with e_s and e_Q denoting the electric charges of the strange and heavy quarks, respectively. The pseudoscalar meson is interpolated by the current

$$J_5 = \bar{s}\, i\gamma_5 Q. \tag{4}$$

For the axial-vector channel, we employ two independent interpolating currents:

$$\begin{aligned} J^A_\mu &= \bar{s}\gamma_\mu\gamma_5 Q, \\ J^B_\mu &= \frac{i}{m_Q + m_s}\bar{s}\sigma_{\mu\nu}\gamma_5 Q\, p'^\nu. \end{aligned} \tag{5}$$

We identify J^A_μ with the $^3P_1$ basis state and J^B_μ with the $^1P_1$ basis state, adopting the same relative phase convention as in Ref. [29]. With this convention, the physical states are defined as

$$\begin{aligned} |A_1\rangle &= \sin\theta\, |^3P_1\rangle + \cos\theta\, |^1P_1\rangle, \\ |A_2\rangle &= \cos\theta\, |^3P_1\rangle - \sin\theta\, |^1P_1\rangle. \end{aligned} \tag{6}$$

Consistent with the state-mixing convention in Eq. (6), we construct the interpolating currents for

the two physical states using the same rotation:

$$\begin{pmatrix} J_\mu^{(1)} \\ J_\mu^{(2)} \end{pmatrix} = \begin{pmatrix} \sin\theta & \cos\theta \\ \cos\theta & -\sin\theta \end{pmatrix} \begin{pmatrix} J_\mu^{A} \\ J_\mu^{B} \end{pmatrix}. \tag{7}$$

Following the standard QCD sum-rule procedure, the correlation function is calculated in two different representations. On the hadronic side, it is expressed in terms of physical hadronic states, whereas on the QCD side it is evaluated in terms of quark and gluon degrees of freedom.

The hadronic representation is obtained by inserting complete sets of states with the same quantum numbers as the interpolating currents and isolating the ground-state contributions. This gives

$$\Pi_{\mu\nu}^{(i)} = \frac{\langle 0|J_\mu^{(i)}|A_i(p',\eta)\rangle\langle A_i(p',\eta)|J_\nu^{\rm em}|P(p)\rangle\langle P(p)|J_5^\dagger|0\rangle}{(m_{A_i}^2 - p'^2)(m_P^2 - p^2)} + \cdots, \tag{8}$$

where the ellipsis denotes contributions from higher resonances and continuum states.

The relevant matrix elements are defined as

$$\langle 0|J_\mu^{(i)}|A_i(p',\eta)\rangle = m_{A_i} f_i \eta_\mu, \tag{9}$$

and

$$\langle P(p)|J_5^\dagger|0\rangle = \frac{m_P^2 f_P}{m_Q + m_s}. \tag{10}$$

Here, f_i and f_P denote the decay constants of the axial-vector meson A_i and the pseudoscalar meson P, respectively.

Lorentz covariance and electromagnetic gauge invariance allow the matrix element to be parametrized as

$$\langle A_i(p',\eta)|J_\nu^{\rm em}|P(p)\rangle = g_i\left[(p\cdot q)\eta_\nu - (\eta\cdot q)p_\nu\right], \tag{11}$$

where

$$q = p' - p,$$

and $g_i(q^2)$ denotes the electric dipole transition form factor. Our aim is to determine this form factor for a real photon, $q^2 = 0$.

After summing over the axial-vector polarization states, Eq. (8) takes the form

$$\Pi_{\mu\nu}^{(i)} = \frac{m_{A_i} f_i g_i m_P^2 f_P}{(m_Q + m_s)(m_{A_i}^2 - p'^2)(m_P^2 - p^2)} \left[(p\cdot q)\left(-g_{\mu\nu} + \frac{p'_\mu p'_\nu}{m_{A_i}^2}\right) - p_\nu\left(-q_\mu + \frac{p'_\mu (p'\cdot q)}{m_{A_i}^2}\right)\right]. \tag{12}$$

After contracting both sides with the photon polarization vector ε^ν, the correlation function can be written as

$$\Pi^{(i)}_{\mu\nu}\varepsilon^\nu = \frac{m_{A_i} f_i g_i m_P^2 f_P}{(m_Q+m_s)(m_{A_i}^2-p'^2)(m_P^2-p^2)}\left\{-\varepsilon_\mu(p\cdot q)+(p\cdot\varepsilon)q_\mu+\cdots\right\}, \tag{13}$$

where the omitted terms correspond to other independent Lorentz structures. We derive the light-cone QCD sum rules for the transition form factors $g_i(q^2=0)$ from the coefficient of the Lorentz structure $(p\cdot\varepsilon)q_\mu-\varepsilon_\mu(p\cdot q)$.

We now turn to the calculation of the correlation function on the QCD side. The QCD representation is evaluated in the deep Euclidean region,

$$p^2 \ll (m_Q+m_s)^2, \qquad p'^2 \ll (m_Q+m_s)^2,$$

where the operator product expansion (OPE) is applicable. To describe the interaction with the external photon, we employ the background-field method, in which the electromagnetic field is treated as a classical external field.

For a real photon with momentum q and polarization vector ε_μ, the electromagnetic field-strength tensor is taken as

$$F_{\mu\nu}(x) = i\left(\varepsilon_\nu q_\mu - \varepsilon_\mu q_\nu\right)e^{iq\cdot x}. \tag{14}$$

Within this formalism, the correlation function in Eq. (2) can be rewritten as

$$\varepsilon^\nu \Pi^{(i)}_{\mu\nu}(p,q) = i\int d^4x\, e^{ip\cdot x}\left\langle 0\left|T\left\{J^{(i)}_\mu(x)J_5^\dagger(0)\right\}\right|0\right\rangle_F, \tag{15}$$

where the subscript F indicates that the vacuum expectation value is evaluated in the presence of the external electromagnetic field.

The correlation function receives two types of contributions. The first corresponds to the perturbative emission of the photon from one of the quark propagators. The second describes the long-distance interaction of the photon with the quark fields and is parameterized in terms of photon light-cone distribution amplitudes (DAs).

Applying Wick's theorem, the correlation function can be expressed in terms of the heavy- and light-quark propagators as

$$\varepsilon^\nu \Pi^{(i)}_{\mu\nu} = i\int d^4x\, e^{ip\cdot x}\mathrm{Tr}\left[\Gamma^{(i)}_\mu S_Q^{ab}(x)\gamma_5 S_s^{ba}(-x)\right]_F, \tag{16}$$

where $\Gamma^{(i)}_\mu$ denotes the Dirac structure associated with the chosen interpolating current, while a and

b are color indices.

It follows from Eq. (16) that the calculation of the correlation function requires the light- and heavy-quark propagators in the presence of external gluonic and electromagnetic background fields. For $j = s, Q$, the corresponding propagator can be written as

$$\begin{aligned} S_j(x) = &\int \frac{d^4k}{(2\pi)^4} e^{-ik\cdot x} \frac{i(\not{k} + m_j)}{k^2 - m_j^2} \\ &- ig_s \int \frac{d^4k}{(2\pi)^4} e^{-ik\cdot x} \int_0^1 du \left[\frac{\not{k} + m_j}{2(m_j^2 - k^2)^2} G^{\mu\nu}(ux)\sigma_{\mu\nu} + \frac{ux_\mu G^{\mu\nu}(ux)\gamma_\nu}{m_j^2 - k^2} \right] \\ &- ie_j \int \frac{d^4k}{(2\pi)^4} e^{-ik\cdot x} \int_0^1 du \left[\frac{\not{k} + m_j}{2(m_j^2 - k^2)^2} F^{\mu\nu}(ux)\sigma_{\mu\nu} + \frac{ux_\mu F^{\mu\nu}(ux)\gamma_\nu}{m_j^2 - k^2} \right] + \cdots , \end{aligned} \tag{17}$$

Here the subscript j denotes either the strange quark (s) or the heavy quark (Q). Furthermore, $G_{\mu\nu}$ and $F_{\mu\nu}$ are the external gluonic and electromagnetic field-strength tensors, respectively, and e_j is the electric charge of the corresponding quark.

The nonperturbative contribution is obtained by replacing the light-quark propagator in Eq. (16) according to

$$S_{\alpha\beta}(x) = -\frac{1}{4} (\Gamma_k)_{\alpha\beta} \; \bar{s}(x)\Gamma_k s(0), \tag{18}$$

where

$$\Gamma_k = \left\{ I,\, \gamma_5,\, \gamma_\mu,\, i\gamma_\mu\gamma_5,\, \frac{\sigma_{\mu\nu}}{\sqrt{2}} \right\} \tag{19}$$

forms a complete basis of Dirac matrices.

This replacement gives rise to the following nonlocal matrix elements:

$$\begin{aligned} &\langle 0|\bar{s}(x)\Gamma_i s(0)|0\rangle_F, \\ &\langle 0|\bar{s}(x)\Gamma_i G_{\mu\nu} s(0)|0\rangle_F, \\ &\langle 0|\bar{s}(x)\Gamma_i F_{\mu\nu} s(0)|0\rangle_F, \end{aligned} \tag{20}$$

which are parameterized in terms of photon light-cone distribution amplitudes of definite twist. In the present analysis, we employ the complete set of two-particle and three-particle photon distribution amplitudes up to twist four. Their explicit expressions are collected in Appendix A.

The OPE representation of the correlation function can therefore be written as the sum of perturbative and nonperturbative contributions:

$$\Pi_K^{\rm OPE} = \Pi_K^{\rm pert} + \Pi_K^{\rm 2p} + \Pi_K^{\rm 3p,g} + \Pi_K^{\rm 3p,\gamma}, \qquad K = A, B, \tag{21}$$

where $\Pi_K^{2\mathrm{p}}$ denotes the contribution of two-particle quark–antiquark photon DAs, while $\Pi_K^{3\mathrm{p},g}$ and $\Pi_K^{3\mathrm{p},\gamma}$ denote the three-particle contributions involving quark–antiquark–gluon and quark–antiquark–photon operators, respectively. The two-particle contribution contains terms of twist two, three, and four. Each term in Eq. (21) refers to the invariant amplitude multiplying the gauge-invariant Lorentz structure

$$(p\cdot\varepsilon)q_\mu - \varepsilon_\mu(p\cdot q).$$

The perturbative contribution is written in terms of spectral densities, and the continuum subtraction is implemented using quark–hadron duality. Its double-Borel-transformed expression is given below in terms of $\rho_A^P(s)$ and $\rho_B^P(s)$. The two- and three-particle photon-DA contributions are transformed separately using the continuum-subtraction prescription specified below.

Since the physical interpolating currents are linear combinations of the two basis currents, the corresponding invariant amplitudes are given by

$$\begin{aligned}\widehat{\Pi}_1 &= \sin\theta\,\widehat{\Pi}_A + \cos\theta\,\widehat{\Pi}_B,\\ \widehat{\Pi}_2 &= \cos\theta\,\widehat{\Pi}_A - \sin\theta\,\widehat{\Pi}_B.\end{aligned} \tag{22}$$

Having determined the coefficient of the selected gauge-invariant Lorentz structure on both the hadronic and OPE sides, we equate the two representations. We then perform double Borel transformations with respect to p'^2 and p^2, which suppress contributions from higher states, and implement the continuum subtraction using quark–hadron duality. Combining the Borel-transformed basis amplitudes with the physical-current rotation yields the sum rules for the transition form factors $g_i(0)$. Before imposing a relation between the two Borel parameters, the matching condition takes the form

$$g_i(0)\,m_{A_i} f_i \frac{m_P^2 f_P}{m_Q + m_s} \exp\left(-\frac{m_{A_i}^2}{M_1^2} - \frac{m_P^2}{M_2^2}\right) = \widehat{\Pi}_i(M_1^2, M_2^2, s_0), \tag{23}$$

It is convenient to introduce the effective Borel parameter and momentum fractions

$$M^2 = \frac{M_1^2 M_2^2}{M_1^2 + M_2^2}, \qquad u_0 = \frac{M_2^2}{M_1^2 + M_2^2}, \qquad a_0 = 1 - u_0. \tag{24}$$

In the numerical analysis, we employ the symmetric double-Borel prescription

$$M_1^2 = M_2^2 = 2M^2, \qquad u_0 = a_0 = \frac{1}{2}. \tag{25}$$

With this choice, Eq. (23) becomes

$$g_i(0)\, m_{A_i} f_i \frac{m_P^2 f_P}{m_Q + m_s} \exp\left[-\frac{m_{A_i}^2 + m_P^2}{2M^2}\right] = \widehat{\Pi}_i(M^2, s_0). \tag{26}$$

After the double Borel transformation, each invariant amplitude is decomposed into perturbative, twist-2, twist-3, twist-4, and three-particle gluonic and electromagnetic contributions:

$$\widehat{\Pi}_{A(B)} = \widehat{\Pi}_{A(B)}^{\text{pert}} + \widehat{\Pi}_{A(B)}^{\text{twist2}} + \widehat{\Pi}_{A(B)}^{\text{twist3}} + \widehat{\Pi}_{A(B)}^{\text{twist4}} + \widehat{\Pi}_{A(B)}^{\text{3p,g}} + \widehat{\Pi}_{A(B)}^{\text{3p,}\gamma}. \tag{27}$$

The perturbative contributions are expressed in terms of the spectral densities $\rho_A(s)$ and $\rho_B(s)$ associated with the axial-vector and tensor basis currents, respectively.

For photon emission from a quark of flavor a, with the other quark denoted by b, we introduce the auxiliary spectral density

$$\begin{aligned}\rho_a(s; m_a, m_b, e_a) = & -\frac{3e_a}{8\pi^2}\left[2m_a \ln\left(\frac{s - m_b^2 + m_a^2 - \sqrt{\lambda(s, m_b^2, m_a^2)}}{s - m_b^2 + m_a^2 + \sqrt{\lambda(s, m_b^2, m_a^2)}}\right)\right. \\ & \left. + (m_b - m_a)\frac{m_b^2 - m_a^2 - s}{s^2}\sqrt{\lambda(s, m_b^2, m_a^2)}\right].\end{aligned} \tag{28}$$

The perturbative spectral density associated with the axial-vector basis current is then

$$\rho_A(s) = \rho_s(s; m_s, m_Q, e_s) - \rho_Q(s; m_Q, m_s, e_Q). \tag{29}$$

For the tensor basis current, the perturbative spectral density is

$$\begin{aligned}\rho_B(s) = -\frac{3}{8\pi^2}\Bigg\{ & e_s\left[\frac{2m_Q m_s}{m_Q + m_s}\ln\left(\frac{s - m_Q^2 + m_s^2 - \sqrt{\lambda(s, m_Q^2, m_s^2)}}{s - m_Q^2 + m_s^2 + \sqrt{\lambda(s, m_Q^2, m_s^2)}}\right)\right. \\ & \left. + \frac{s}{m_Q + m_s}\frac{m_Q^2 - m_s^2 - s}{s^2}\sqrt{\lambda(s, m_Q^2, m_s^2)}\right] \\ & - e_Q\left[\frac{2\left(2m_Q^2 - m_Q m_s\right)}{m_Q + m_s}\ln\left(\frac{s - m_s^2 + m_Q^2 - \sqrt{\lambda(s, m_s^2, m_Q^2)}}{s - m_s^2 + m_Q^2 + \sqrt{\lambda(s, m_s^2, m_Q^2)}}\right)\right. \\ & \left. - \frac{s}{m_Q + m_s}\frac{m_s^2 - m_Q^2 - s}{s^2}\sqrt{\lambda(s, m_s^2, m_Q^2)}\right]\Bigg\},\end{aligned} \tag{30}$$

where

$$\lambda(a, b, c) = (a - b - c)^2 - 4bc, \tag{31}$$

is the Källén function.

The corresponding continuum-subtracted and Borel-transformed perturbative amplitudes are

$$\widehat{\Pi}_K^{\rm pert}(M^2,s_0)=\int_{(m_Q+m_s)^2}^{s_0} ds\, e^{-s/M^2}\rho_K(s), \qquad K=A,B. \tag{32}$$

For the photon-DA contributions, the continuum subtraction is implemented term by term. The twist-2, twist-3, two-particle twist-4, gluonic three-particle, and electromagnetic three-particle contributions are all multiplied by

$$e^{-m_Q^2/M^2}-e^{-s_0/M^2}.$$

The finite strange-quark mass does not modify this soft-sector continuum factor. It is retained in the perturbative threshold $(m_Q+m_s)^2$, the spectral densities, the current normalization, and the mass-dependent coefficient functions.

With this prescription, the two-particle contributions associated with the axial-vector basis current are

$$\begin{aligned}
\widehat{\Pi}_A^{\rm twist2} &= e_s f_{\gamma,s}^{\perp}\left(e^{-m_Q^2/M^2}-e^{-s_0/M^2}\right)M^2\phi_\gamma(u_0),\\
\widehat{\Pi}_A^{\rm twist3} &= e_s f_{3\gamma} m_Q\left(e^{-m_Q^2/M^2}-e^{-s_0/M^2}\right)\psi^{(V)}(u_0),\\
\widehat{\Pi}_A^{\rm twist4} &= \frac{e_s\langle\bar{s}s\rangle}{4}\left(e^{-m_Q^2/M^2}-e^{-s_0/M^2}\right)\left[\mathcal{A}(u_0)+2\mathcal{B}(u_0)\right]\left(1+\frac{m_Q^2}{M^2}\right).
\end{aligned} \tag{33}$$

Here, $\phi_\gamma(u)$, $\psi^{(V)}(u)$, $\mathcal{A}(u)$, and $\mathcal{B}(u)$ denote the two-particle photon DAs of twist two, three, and four, respectively. Their explicit expressions and normalization conventions are collected in Appendix A.

For the tensor basis current, the twist-3 contribution is

$$\widehat{\Pi}_B^{\rm twist3}=\frac{e_s f_{3\gamma}}{m_Q+m_s}\left(e^{-m_Q^2/M^2}-e^{-s_0/M^2}\right)\left[m_Q^2\psi^{(V)}(u_0)-\frac{M^2}{2}\frac{d}{da}\left\{(1-a)\psi^{(V)}(a)\right\}\Big|_{a=u_0}\right]. \tag{34}$$

The twist-2 and twist-4 contributions associated with the tensor basis current are related to the corresponding axial-vector-current contributions through

$$\widehat{\Pi}_B^{\rm twist2}=\frac{m_Q}{m_Q+m_s}\widehat{\Pi}_A^{\rm twist2}, \qquad \widehat{\Pi}_B^{\rm twist4}=\frac{m_Q}{m_Q+m_s}\widehat{\Pi}_A^{\rm twist4}. \tag{35}$$

Explicit expressions for $\widehat{\Pi}_K^{\rm 3p,g}$ and $\widehat{\Pi}_K^{\rm 3p,\gamma}$, with $K=A,B$, are collected in Appendix B.

The sum rule in Eq. (23) depends on the decay constants of the physical axial-vector mesons. These quantities are determined from the two-point correlation function

$$\Pi^{(i)}_{\mu\nu}(p) = i \int d^4x\, e^{ip\cdot x} \langle 0|T\left\{J^{(i)}_{\mu}(x) J^{(i)\dagger}_{\nu}(0)\right\}|0\rangle, \tag{36}$$

where the interpolating currents $J^{(i)}_{\mu}$ are defined in Eq. (7).
The resulting sum rules for the decay constants are

$$f_1^2 m_{A_1}^2 e^{-m_{A_1}^2/M^2} = \sin^2\theta\, \widehat{\Pi}_{AA} + 2\sin\theta\cos\theta\, \widehat{\Pi}_{AB} + \cos^2\theta\, \widehat{\Pi}_{BB}, \tag{37}$$

$$f_2^2 m_{A_2}^2 e^{-m_{A_2}^2/M^2} = \cos^2\theta\, \widehat{\Pi}_{AA} - 2\sin\theta\cos\theta\, \widehat{\Pi}_{AB} + \sin^2\theta\, \widehat{\Pi}_{BB}. \tag{38}$$

Here, $\widehat{\Pi}_{AA}$, $\widehat{\Pi}_{AB}$, and $\widehat{\Pi}_{BB}$ denote the Borel-transformed invariant amplitudes obtained from the corresponding two-point correlation functions, and their expressions are presented in Appendix C.
For completeness, we briefly summarize the determination of the mixing angles in Ref. [29]. The calculation employed the same normalized axial-vector and tensor currents used here.[1]
Within the two-state description adopted in that analysis, the mixing angle is determined by requiring the transverse two-point correlation matrix to be diagonal in the rotated current basis. Writing

$$\widehat{\mathbf{\Pi}} = \begin{pmatrix} \widehat{\Pi}_{AA} & \widehat{\Pi}_{AB} \\ \widehat{\Pi}_{AB} & \widehat{\Pi}_{BB} \end{pmatrix}, \qquad R(\theta) = \begin{pmatrix} \sin\theta & \cos\theta \\ \cos\theta & -\sin\theta \end{pmatrix}, \tag{39}$$

where $\widehat{\Pi}_{AB} = \widehat{\Pi}_{BA}$, the diagonalization condition is

$$R(\theta)\widehat{\mathbf{\Pi}} R^T(\theta) = \begin{pmatrix} \widehat{\Pi}_{11} & 0 \\ 0 & \widehat{\Pi}_{22} \end{pmatrix}. \tag{40}$$

Here, $\widehat{\Pi}_{11}$ and $\widehat{\Pi}_{22}$ denote the diagonal two-point amplitudes in the rotated basis. All amplitudes are evaluated after Borel transformation and continuum subtraction, using a common two-point Borel parameter and continuum prescription. The vanishing off-diagonal element gives

$$\frac{1}{2}\sin 2\theta \left(\widehat{\Pi}_{AA} - \widehat{\Pi}_{BB}\right) + \cos 2\theta\, \widehat{\Pi}_{AB} = 0. \tag{41}$$

[1] The factor $1/(m_Q + m_s)$ in the tensor current was inadvertently omitted from the printed current definition in Ref. [29]; it was included in the calculations underlying the reported mixing angles.

The role of the physical diagonal correlators can be seen explicitly by applying the inverse rotation:

$$\widehat{\Pi}_{AA} - \widehat{\Pi}_{BB} = -\cos 2\theta \left(\widehat{\Pi}_{11} - \widehat{\Pi}_{22}\right),$$
$$2\widehat{\Pi}_{AB} = \sin 2\theta \left(\widehat{\Pi}_{11} - \widehat{\Pi}_{22}\right). \tag{42}$$

The two physical diagonal correlators may contain different pole masses and residues. Their difference is not assumed to vanish; it appears as a common factor that cancels when the ratio of these relations is taken. Thus, wherever the ratio is defined, the mixing-angle sum rule becomes

$$\tan 2\theta = -\frac{2\widehat{\Pi}_{AB}}{\widehat{\Pi}_{AA} - \widehat{\Pi}_{BB}}. \tag{43}$$

The angle branch is fixed by the adopted current convention and the assignments of the two physical states. In Ref. [29], the basis correlators were calculated using the operator product expansion, and the extracted angle was studied over the accepted Borel and continuum-threshold regions. The resulting strange-sector mixing angles are quoted in the numerical analysis below.

Substituting the residues obtained from the two-point sum rules into the symmetric transition sum rule, and using the physical-current rotation, we obtain the form factors

$$g_1(0) = \frac{m_Q + m_s}{m_{A_1} f_1 m_P^2 f_P} \exp\left(\frac{m_{A_1}^2 + m_P^2}{2M^2}\right) \left[\sin\theta\, \widehat{\Pi}_A(M^2, s_0) + \cos\theta\, \widehat{\Pi}_B(M^2, s_0)\right],$$
$$g_2(0) = \frac{m_Q + m_s}{m_{A_2} f_2 m_P^2 f_P} \exp\left(\frac{m_{A_2}^2 + m_P^2}{2M^2}\right) \left[\cos\theta\, \widehat{\Pi}_A(M^2, s_0) - \sin\theta\, \widehat{\Pi}_B(M^2, s_0)\right]. \tag{44}$$

For each transition, the invariant amplitudes in Eq. (44) are evaluated using the corresponding heavy-quark flavor, continuum threshold, and Borel window. The transition form factors obtained from the light-cone QCD sum rules can be used to determine the radiative decay widths of the physical axial-vector mesons. The partial decay width of the $A_i \to P\gamma$ process is

$$\Gamma(A_i \to P\gamma) = \frac{\alpha_{\rm em}}{3} g_i^2(0) \left(\frac{m_{A_i}^2 - m_P^2}{2m_{A_i}}\right)^3, \tag{45}$$

where $\alpha_{\rm em}$ is the electromagnetic fine-structure constant.

Once the transition form factors are determined, Eq. (45) gives the corresponding radiative decay widths. The numerical analysis is presented in the following section.

TABLE I: The values of the input parameters used in the numerical analysis. The condensate inputs and photon-DA normalization are quoted at $\mu = 1$ GeV.

Parameter	Value
$\overline{m}_c(\overline{m}_c)$, $\overline{m}_b(\overline{m}_b)$, $\overline{m}_s(2\text{ GeV})$	1.27(2), 4.18(3), 0.093(11) GeV [32]
$m_{D_{s1}(2460)}$, $m_{D_{s1}(2536)}$, m_{D_s}	2.4595(6), 2.53512(6), 1.96835(7) GeV [32]
$m_{B_{s1}(5750)}$, $m_{B_{s1}(5830)}$, m_{B_s}	5.750(26) GeV [9], 5.82873(20), 5.36692(10) GeV [32]
f_{D_s}, f_{B_s}	0.2499(5), 0.2303(13) GeV [33]
$\langle\bar{q}q\rangle$	$-(0.240 \pm 0.010)^3$ GeV^3 [31]
$\langle\bar{s}s\rangle/\langle\bar{q}q\rangle$	0.80 ± 0.10 [25]
$m_0^2 \equiv \langle\bar{s}g_s\sigma\cdot Gs\rangle/\langle\bar{s}s\rangle$	0.80 ± 0.20 GeV^2 [25]
$f^{\perp}_{\gamma,s} \equiv \chi_s\langle\bar{s}s\rangle$	$-55.1(1.9)$ MeV [30]

III. NUMERICAL ANALYSIS

The external numerical inputs required for the analysis, including the quark and meson masses, pseudoscalar decay constants, condensates, and the leading-twist photon-DA normalization, are collected in Table I. The remaining photon-DA parameters are listed in Table V. For the leading-twist photon DA, we use the recent lattice determination $f^{\perp}_{\gamma,s}$ [30], which provides a more precise input than the conventional susceptibility parameter χ_s [31].

For the numerical analysis, we use the mixing angles obtained in Ref. [29],

$$\theta_{D_s} = 26.6^\circ \pm 0.6^\circ, \qquad \theta_{B_s} = 38.5^\circ \pm 0.1^\circ. \tag{46}$$

Using these values in the two-point sum rules in Eqs. (37) and (38), we obtain the physical axial-vector residues $f_1^{(D_s)} = 405^{+25}_{-26}$ MeV for $D_{s1}(2460)$, $f_2^{(D_s)} = 168^{+7}_{-8}$ MeV for $D_{s1}(2536)$, $f_1^{(B_s)} = 536^{+23}_{-26}$ MeV for $B_{s1}(5750)$, and $f_2^{(B_s)} = 89^{+4}_{-4}$ MeV for $B_{s1}(5830)$. As a cross-check, at $\theta = 90^\circ$, the first physical current reduces to the unmixed axial-vector current, and the corresponding results are compatible with those reported in [34]. The mixing-angle dependence of the decay constants is shown in Fig. 1 for both heavy-strange sectors. Over the displayed angular range, f_1 varies relatively weakly, whereas f_2 exhibits a more pronounced dependence, particularly in the bottom-strange sector. These curves illustrate the angular dependence at fixed representative inputs; the numerical estimates quoted above also include the propagated input uncertainties.

Because the initial and final mesons in each channel are relatively close in mass, we employ the symmetric Borel prescription $M_1^2 = M_2^2 = 2M^2$. The working regions are selected by requiring stability of the transition form factors, convergence of the light-cone expansion, and sufficient suppression of higher-state and continuum contributions. The adopted intervals are listed in Table II.

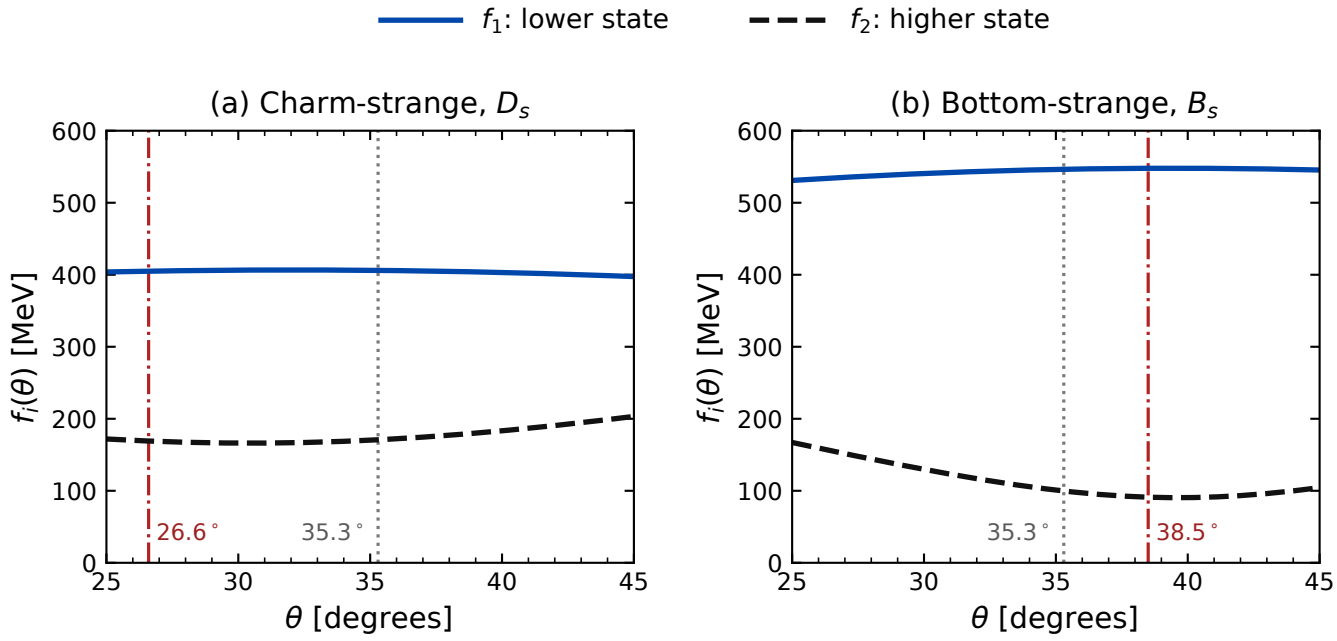

FIG. 1: Mixing-angle dependence of the decay constants $f_1(\theta)$ and $f_2(\theta)$ in the (a) charm-strange and (b) bottom-strange sectors. The solid blue curves correspond to the lower states $D_{s1}(2460)$ and the expected $B_{s1}(5750)$, while the dashed black curves correspond to the higher states $D_{s1}(2536)$ and $B_{s1}(5830)$. The red dash-dotted vertical lines indicate the adopted central mixing angles, $\theta_{D_s} = 26.6^\circ$ and $\theta_{B_s} = 38.5^\circ$; the gray dotted lines mark the heavy-quark-limit reference value 35.3°. Both decay constants are recalculated at each angle, with the QCD inputs, two-point Borel parameters, and continuum thresholds fixed to the same representative choices used in the angular scan of the radiative widths.

TABLE II: Borel windows and continuum-threshold intervals used in the numerical analysis.

Channel	M^2 (GeV^2)	s_0 (GeV^2)
$D_{s1}(2460) \to D_s\gamma$	3.0–4.5	8.5–9.5
$D_{s1}(2536) \to D_s\gamma$	3.0–4.5	9.0–10.0
$B_{s1}(5750) \to B_s\gamma$	10.0–14.0	39.0–41.0
$B_{s1}(5830) \to B_s\gamma$	10.0–14.0	40.0–42.0

The heavy-quark-limit value $\theta = 35.3^\circ$ is retained as a reference.

Figure 2 shows the dependence of the transition form factors on M^2 at fixed values of s_0 and at the mixing angles given in Eq. (46). The residual variation over the full M^2 and s_0 intervals is included in the uncertainty analysis.

The zero crossing of $g_2(0)$ in the $D_{s1}(2536) \to D_s\gamma$ panel of Fig. 2 is a consequence of destructive interference between the two basis-current contributions. The individual contributions vary smoothly with M^2 and s_0, but enter the amplitude with opposite signs and comparable magnitudes. Their cancellation therefore leads to a zero whose position changes with the variation of the input

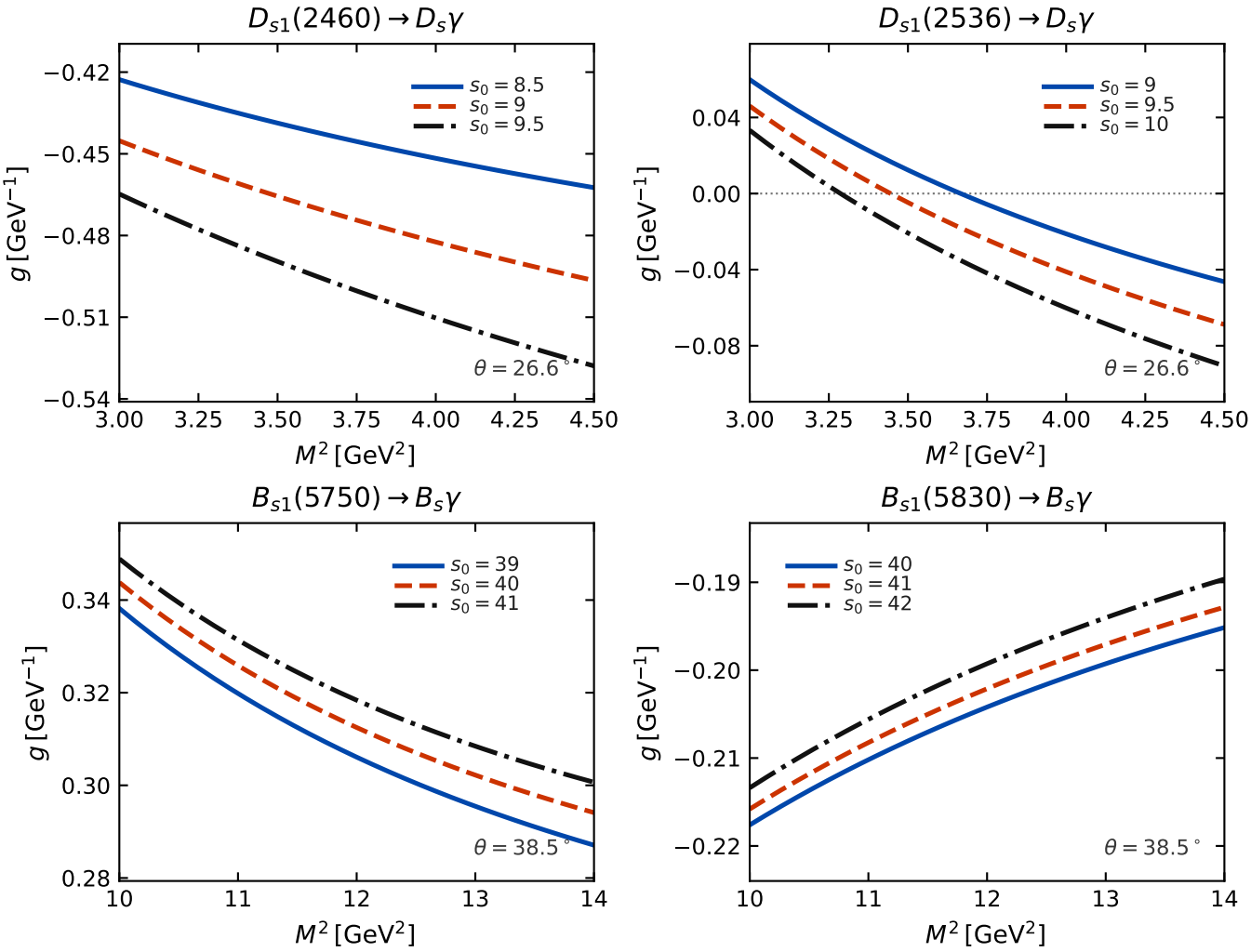

FIG. 2: Dependence of the transition form factors $g(0)$ on the Borel parameter M^2 at fixed values of the continuum threshold s_0. The panels show $D_{s1}(2460) \to D_s\gamma$, $D_{s1}(2536) \to D_s\gamma$, $B_{s1}(5750) \to B_s\gamma$, and $B_{s1}(5830) \to B_s\gamma$.

parameters.

The transition amplitudes are linear combinations of the two basis-current contributions. When these contributions have opposite signs, their destructive interference can produce a clear minimum in the radiative width as the mixing angle is varied. This behavior is illustrated in Fig. 3. The remaining inputs are fixed at their central values in this figure; the curves therefore illustrate only the mixing-angle dependence and should not be identified with the final results in Table III.

The uncertainties are obtained by varying the QCD and hadronic inputs within their quoted ranges, together with the Borel parameter, continuum threshold, and mixing angles. We quote the median values and the corresponding central 68% intervals. The resulting form factors and radiative widths are given in Table III.

The most notable feature is the strong suppression of $D_{s1}(2536) \to D_s\gamma$ relative to $D_{s1}(2460) \to D_s\gamma$. As seen from Figs. 2 and 3, this behavior originates from destructive interference between the two

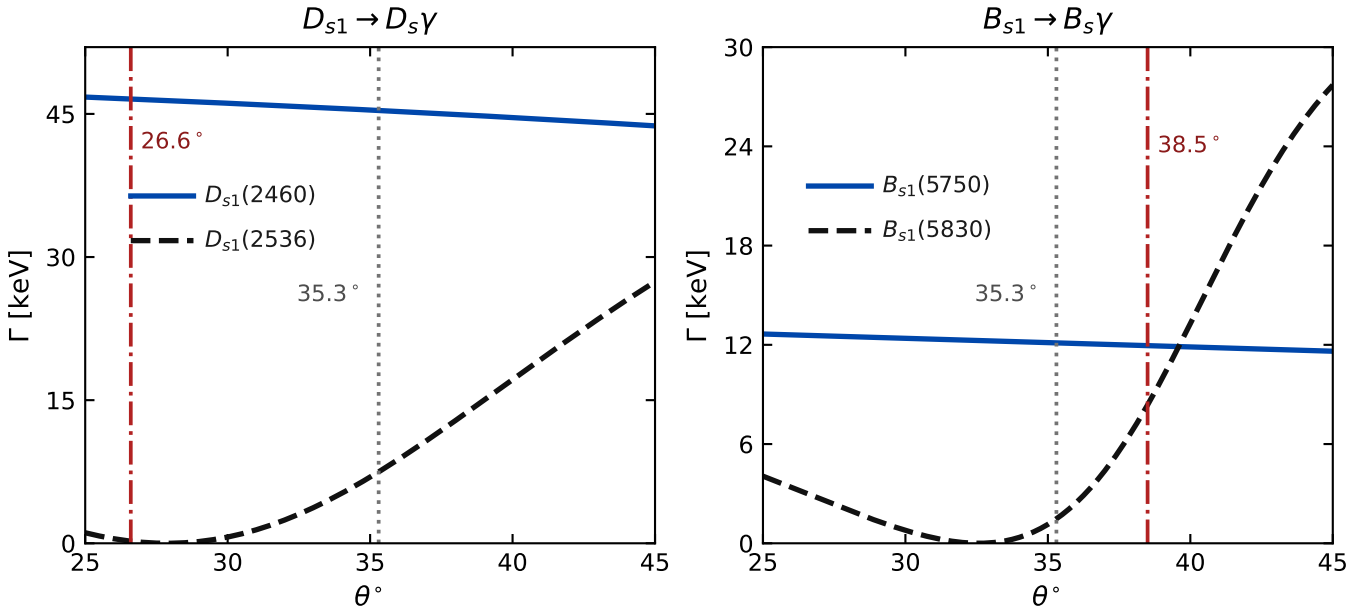

FIG. 3: Dependence of the radiative decay widths on the mixing angle. The charm-strange and bottom-strange channels are shown in the left and right panels, respectively. The mixing angles $\theta_{D_s} = 26.6°$ and $\theta_{B_s} = 38.5°$ are indicated in the corresponding panels, while $\theta = 35.3°$ denotes the heavy-quark-limit reference value. The curves are evaluated with the remaining input parameters fixed at their central values and are shown only to illustrate the mixing-angle dependence; the final results are given in Table III.

basis-current contributions in the higher charm-strange state, while the corresponding contributions add constructively for $D_{s1}(2460)$. The calculation therefore reproduces within the light-cone QCD sum-rule framework the cancellation mechanism discussed in Refs. [27, 28].

In the bottom-strange sector, the two radiative widths are comparable within their uncertainties. The mixing-angle dependence of the higher state remains significant, but the value of θ_{B_s} used in the numerical analysis lies away from the strongest cancellation region. We emphasize that $B_{s1}(5750)$ is not an experimentally established state and is included as the theoretically expected lower $J^P = 1^+$ bottom-strange configuration.

The width distributions shown in Fig. 4 reflect the different sensitivities of the four channels to the input parameters. In particular, the distribution for $D_{s1}(2536) \to D_s\gamma$ is strongly asymmetric, consistent with its cancellation-sensitive amplitude.

Table IV compares our results with selected theoretical predictions available in the literature. The predictions span a broad range, reflecting differences in the treatment of the axial-vector states, their mixing, and the radiative transition dynamics. These differences become particularly important for channels in which the two mixed-state contributions partially cancel. Despite the substantial variation in the absolute decay widths, the suppression of the higher charm-strange state relative to the $D_{s1}(2460)$ is also found in Ref. [28].

TABLE III: The central values are medians, and the uncertainties denote the central 68% propagated uncertainty intervals.

Channel	$\lvert g_i\rvert$ (GeV^{-1})	Γ (keV)
$D_{s1}(2460) \to D_s\gamma$	$0.47^{+0.05}_{-0.05}$	$45.9^{+10.1}_{-8.5}$
$D_{s1}(2536) \to D_s\gamma$	$0.04^{+0.04}_{-0.03}$	$0.55^{+1.73}_{-0.48}$
$B_{s1}(5750) \to B_s\gamma$	$0.32^{+0.02}_{-0.02}$	$12.0^{+2.9}_{-2.3}$
$B_{s1}(5830) \to B_s\gamma$	$0.20^{+0.03}_{-0.02}$	$8.9^{+2.4}_{-2.0}$

TABLE IV: Comparison with selected theoretical predictions. All widths are given in keV. The values from Ref. [35] are obtained from the corresponding radiative couplings using Eq. (45).

Approach / Ref.	$D_{s1}(2460) \to D_s\gamma$	$D_{s1}(2536) \to D_s\gamma$	$B_{s1}(5750) \to B_s\gamma$	$B_{s1}(5830) \to B_s\gamma$
LCSR [25]	19–29	–	–	–
VMD [21, 22]	3.3	–	–	–
Lattice-informed VMD/HQ scaling [35]	11.1 ± 1.9	–	–	4.2 ± 0.7
Relativized quark model [13, 36]	6.2	15	–	–
Relativistic quark model [12]	10.3–17.2	25.2–31.1	–	–
Potential model [18]	13.2	61.2	–	–
Potential model [16]	12.8	54.5	–	–
Relativistic quark model [15]	3.53–3.61	18.18–18.85	–	–
Unquenched coupled-channel model [20]	0.98–6.99	–	2.42–37.27	–
Heavy-quark framework [37]	–	1.6 ± 2.3	–	–
Chiral quark model [19]	–	–	37	27
Relativized quark model [14]	–	–	70.6	47.8
Constituent quark model [17]	–	–	97.7	56.6
Potential model [28]	297	12	47	28
This work (LCSR)	$45.9^{+10.1}_{-8.5}$	$0.55^{+1.73}_{-0.48}$	$12.0^{+2.9}_{-2.3}$	$8.9^{+2.4}_{-2.0}$

Finally, combining our prediction for $D_{s1}(2536) \to D_s\gamma$ with the measured total width gives

$$\mathcal{B}[D_{s1}(2536) \to D_s\gamma] = \left(0.60^{+1.86}_{-0.52}\right) \times 10^{-3}, \tag{47}$$

where the uncertainty is dominated by that of the radiative amplitude. The small branching fraction makes the experimental observation of this decay mode challenging.

IV. CONCLUSION

We have studied the radiative decays $D_{s1}(2460) \to D_s\gamma$, $D_{s1}(2536) \to D_s\gamma$, $B_{s1}(5750) \to B_s\gamma$, and $B_{s1}(5830) \to B_s\gamma$ within light-cone QCD sum rules, taking into account the ${}^1P_1$–${}^3P_1$ mixing of the physical axial-vector states. The decay constants are determined consistently from two-point sum rules and used in the transition analysis. To our knowledge, the $D_{s1}(2536) \to D_s\gamma$ mode is studied

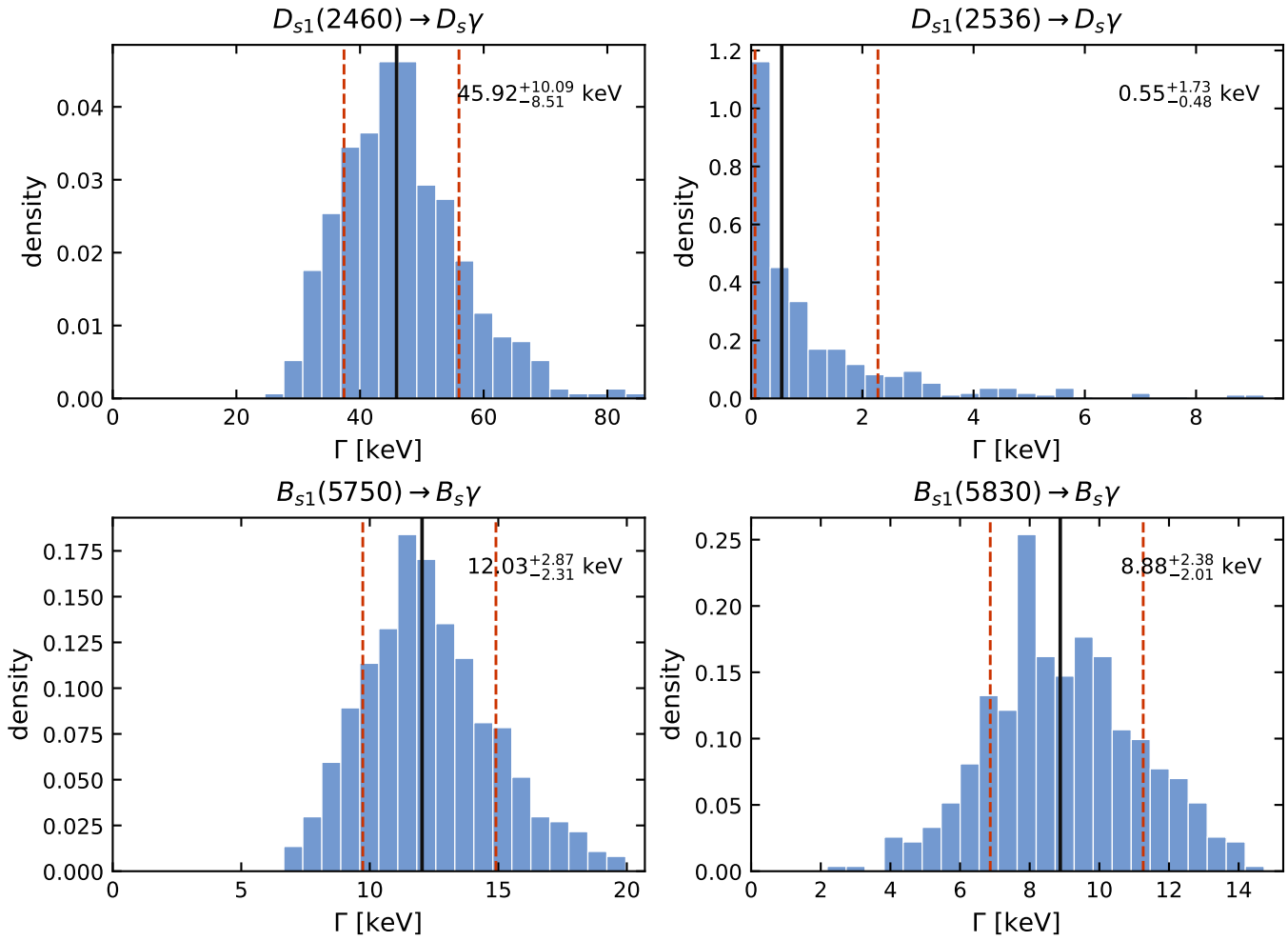

FIG. 4: Monte Carlo distributions of the radiative decay widths obtained with the lattice-normalized photon input and the mixing angles $\theta_{D_s} = 26.6^\circ \pm 0.6^\circ$ and $\theta_{B_s} = 38.5^\circ \pm 0.1^\circ$. The remaining QCD and hadronic inputs, together with the Borel parameter, continuum threshold, and two-point residue samples, are varied as in the final uncertainty analysis. Solid vertical lines indicate the medians, while dashed lines delimit the 16th–84th percentile intervals.

here for the first time within the light-cone QCD sum-rule framework.

The main result is the strong suppression of $D_{s1}(2536) \to D_s\gamma$ relative to $D_{s1}(2460) \to D_s\gamma$. The two basis-current contributions interfere constructively for the lower state, whereas they nearly cancel for the higher state. The light-cone QCD sum-rule calculation therefore provides an independent realization of the interference mechanism discussed in Refs. [27, 28]. The resulting $D_{s1}(2536) \to D_s\gamma$ width is at the sub-keV level, corresponding to a branching fraction of order 10^{-3}. This strong suppression is consistent with the present difficulty of observing this radiative mode experimentally. The bottom-strange system shows a different pattern. At the mixing angle used in the numerical analysis, the predicted widths of $B_{s1}(5750) \to B_s\gamma$ and $B_{s1}(5830) \to B_s\gamma$ are comparable within their uncertainties, although the higher-state amplitude remains sensitive to the interference of the two basis components. The $B_{s1}(5750)$ state is included as the theoretically expected lower $J^P =$

1^+ bottom-strange configuration and should not be interpreted as an experimentally established resonance.

Our results show that the radiative decays of heavy-strange axial-vector mesons are strongly influenced by the relative magnitude and sign of the $^1P_1$ and $^3P_1$ contributions and, consequently, by the mixing angle. This sensitivity is particularly strong for channels close to a cancellation and also helps explain the wide spread among existing theoretical predictions. Future measurements of $D_{s1}(2536) \to D_s\gamma$ and $B_{s1}(5830) \to B_s\gamma$, together with improved information on other radiative modes, could therefore provide constraints on the $^1P_1$–$^3P_1$ mixing angles and discriminate among different descriptions of the physical axial-vector states. Searches for the lower bottom-strange 1^+ state would provide an additional test of this picture.

Appendix A: Photon distribution amplitudes

Photon DAs provide the principal nonperturbative input to the light-cone QCD sum rules. We use the photon DAs required in the present calculation up to twist four, following Ref. [38].

The leading-twist two-particle photon DA is taken in the asymptotic form

$$\phi_\gamma(u) = 6u\bar{u}. \tag{A1}$$

The twist-3 vector DA is

$$\begin{aligned}\psi^{(V)}(u) = & -20u\bar{u}(2u-1) \\ & + \frac{15}{16}\left(\omega_\gamma^A - 3\omega_\gamma^V\right) u\bar{u}(2u-1)\left[7(2u-1)^2 - 3\right].\end{aligned} \tag{A2}$$

For the twist-4 two-particle DAs, we use

$$\begin{aligned}\mathcal{A}(u) = & \, 40u^2\bar{u}^2\left(3\kappa - \kappa^+ + 1\right) \\ & + 8\left(\zeta_2^+ - 3\zeta_2\right)\Big[u\bar{u}(2+13u\bar{u}) \\ & \qquad + 2u^3(10-15u+6u^2)\ln u + 2\bar{u}^3(10-15\bar{u}+6\bar{u}^2)\ln\bar{u}\Big],\end{aligned} \tag{A3}$$

$$\mathcal{B}(u) = 40(1+3\kappa^+)\int_0^u d\alpha\,(u-\alpha)\left[-\frac{1}{2} + \frac{3}{2}(2\alpha-1)^2\right]. \tag{A4}$$

The gluonic twist-4 three-particle photon DAs are

$$\mathcal{S}(\alpha_i) = 30\alpha_g^2\Big[(\kappa+\kappa^+)(1-\alpha_g)$$

TABLE V: Photon distribution-amplitude parameters at the renormalization scale $\mu = 1$ GeV. The twist-3 and twist-4 parameters are adopted from Refs. [31, 38]; in particular, the values $\kappa = 0.15$ and $\kappa^+ = -0.05$ are taken from Ref. [31].

Parameter	Value
$f_{\gamma,s}^{\perp}$	$-55.1(1.9)$ MeV [30]
$f_{3\gamma}$	$-(4.0 \pm 2.0) \times 10^{-3}$ GeV^2
ω_γ^A	-2.1 ± 1.0
ω_γ^V	3.8 ± 1.8
κ	0.15
κ^+	-0.05
ζ_1	0.40
ζ_1^+	0
ζ_2	0.30
ζ_2^+	0

$$
\begin{aligned}
&+ (\zeta_1 + \zeta_1^+)(1-\alpha_g)(1-2\alpha_g) \\
&+ \zeta_2 \left\{3(\alpha_{\bar s} - \alpha_s)^2 - \alpha_g(1-\alpha_g)\right\}\Big], \qquad \text{(A5)}
\end{aligned}
$$

$$
\begin{aligned}
\widetilde{\mathcal{S}}(\alpha_i) = &-30\alpha_g^2\Big[(\kappa - \kappa^+)(1-\alpha_g) \\
&+ (\zeta_1 - \zeta_1^+)(1-\alpha_g)(1-2\alpha_g) \\
&+ \zeta_2 \left\{3(\alpha_{\bar s} - \alpha_s)^2 - \alpha_g(1-\alpha_g)\right\}\Big], \qquad \text{(A6)}
\end{aligned}
$$

$$
\mathcal{T}_1(\alpha_i) = -120(3\zeta_2 + \zeta_2^+)(\alpha_{\bar s} - \alpha_s)\alpha_{\bar s}\alpha_s\alpha_g, \qquad \text{(A7)}
$$

$$
\begin{aligned}
\mathcal{T}_2(\alpha_i) = 30\alpha_g^2(\alpha_{\bar s} - \alpha_s)\Big[&\kappa - \kappa^+ \\
&+ (\zeta_1 - \zeta_1^+)(1-2\alpha_g) + \zeta_2(3-4\alpha_g)\Big], \qquad \text{(A8)}
\end{aligned}
$$

$$
\mathcal{T}_3(\alpha_i) = -120(3\zeta_2 - \zeta_2^+)(\alpha_{\bar s} - \alpha_s)\alpha_{\bar s}\alpha_s\alpha_g, \qquad \text{(A9)}
$$

$$
\begin{aligned}
\mathcal{T}_4(\alpha_i) = 30\alpha_g^2(\alpha_{\bar s} - \alpha_s)\Big[&\kappa + \kappa^+ \\
&+ (\zeta_1 + \zeta_1^+)(1-2\alpha_g) + \zeta_2(3-4\alpha_g)\Big]. \qquad \text{(A10)}
\end{aligned}
$$

In addition to these gluonic distributions, the background-field expansion contains the electromagnetic three-particle photon DAs $\mathcal{S}_\gamma$ and $\mathcal{T}_4^\gamma$. Their explicit forms are

$$
\mathcal{S}_\gamma(\alpha_i) = 60\alpha_g^2\,(\alpha_s + \alpha_{\bar s})\,[4 - 7\,(\alpha_s + \alpha_{\bar s})]\,, \qquad \text{(A11)}
$$

$$
\mathcal{T}_4^\gamma(\alpha_i) = 60\alpha_g^2\,(\alpha_{\bar s} - \alpha_s)\,[4 - 7\,(\alpha_s + \alpha_{\bar s})]\,. \qquad \text{(A12)}
$$

The photon-DA parameters used in the numerical analysis are summarized in Table V.

Appendix B: Explicit expressions for the three-particle invariant amplitudes $\widehat{\Pi}^{3\mathrm{p},\gamma}_{A(B)}$ and $\widehat{\Pi}^{3\mathrm{p,g}}_{A(B)}$

For the electromagnetic three-particle photon DAs, we define

$$\begin{aligned}\mathcal{F}^{\gamma}_{A}(\alpha_i,v) &= (1-2v)\mathcal{S}_{\gamma}(\alpha_i) - \mathcal{T}^{\gamma}_{4}(\alpha_i),\\ \mathcal{F}^{\gamma}_{B}(\alpha_i) &= \frac{m_Q}{m_Q+m_s}\left[\mathcal{S}_{\gamma}(\alpha_i) - \mathcal{T}^{\gamma}_{4}(\alpha_i)\right].\end{aligned} \tag{B1}$$

The corresponding three-particle electromagnetic contributions are

$$\begin{aligned}\widehat{\Pi}^{3\mathrm{p},\gamma}_{A} &= e_Q\langle\bar{s}s\rangle\left(e^{-m_Q^2/M^2}-e^{-s_0/M^2}\right)\left\{\mathcal{J}_{\bar{u}_0}[\mathcal{F}^{\gamma}_{A}] + 2\frac{d}{dz}\mathcal{L}_z[\mathcal{T}^{\gamma}_{4}]\Big|_{z=\bar{u}_0}\right\},\\ \widehat{\Pi}^{3\mathrm{p},\gamma}_{B} &= e_Q\langle\bar{s}s\rangle\left(e^{-m_Q^2/M^2}-e^{-s_0/M^2}\right)\frac{m_Q}{m_Q+m_s}\left\{\mathcal{J}_{\bar{u}_0}[\mathcal{S}_{\gamma}-\mathcal{T}^{\gamma}_{4}] + 2\frac{d}{dz}\mathcal{L}_z[\mathcal{T}^{\gamma}_{4}]\Big|_{z=\bar{u}_0}\right\},\end{aligned} \tag{B2}$$

where

$$\begin{aligned}\mathcal{J}_z[X] &= \int_0^z dv\int_0^{(1-z)/(1-v)} d\alpha_g\, X(1-z-(1-v)\alpha_g,\, z-v\alpha_g,\, \alpha_g,\, v)\\ &\quad+\int_z^1 dv\int_0^{z/v} d\alpha_g\, X(1-z-(1-v)\alpha_g,\, z-v\alpha_g,\, \alpha_g,\, v)\,,\end{aligned} \tag{B3}$$

$$\begin{aligned}\mathcal{L}_z[X] &= \int_0^{1-z} d\alpha_s\int_0^{\alpha_s} d\alpha'_s\int_0^z d\alpha_{\bar{s}}\,\frac{z-\alpha_{\bar{s}}}{(1-\alpha_s-\alpha_{\bar{s}})^2}X(\alpha_s,\,\alpha_{\bar{s}},\,1-\alpha_s-\alpha_{\bar{s}})\\ &\quad-\int_0^{1-z} d\alpha_s\int_z^{1-\alpha_s} d\beta\,\frac{z}{\beta^2}\int_0^{\beta} d\alpha'_{\bar{s}}\,X(\alpha_s,\,\alpha'_{\bar{s}},\,1-\alpha_s-\alpha'_{\bar{s}})\\ &\quad+\int_0^{1-z} d\alpha_s\,\frac{z}{(1-\alpha_s)^2}\int_0^{\alpha_s} d\alpha'_s\int_0^{1-\alpha'_s} d\alpha_{\bar{s}}\,X(\alpha'_s,\,\alpha_{\bar{s}},\,1-\alpha'_s-\alpha_{\bar{s}}).\end{aligned} \tag{B4}$$

The gluonic three-particle contributions are determined by the functions

$$\begin{aligned}\mathcal{F}^{g}_{A,\sigma}(\alpha_i) &= \mathcal{S}+\widetilde{\mathcal{S}}-\mathcal{T}_1-\mathcal{T}_2+\mathcal{T}_3+\mathcal{T}_4,\\ \mathcal{F}^{g}_{A}(\alpha_i,v) &= \mathcal{F}^{g}_{A,\sigma}(\alpha_i)+2v\left(-\mathcal{S}-\mathcal{T}_3+\mathcal{T}_2\right),\\ \mathcal{F}^{g}_{B}(\alpha_i) &= \frac{m_Q}{m_Q+m_s}\mathcal{F}^{g}_{A,\sigma}(\alpha_i).\end{aligned} \tag{B5}$$

The corresponding invariant amplitudes are

$$\begin{aligned}\widehat{\Pi}^{3\mathrm{p,g}}_{A} &= e_s\langle\bar{s}s\rangle\left(e^{-m_Q^2/M^2}-e^{-s_0/M^2}\right)\mathcal{J}_{\bar{u}_0}[\mathcal{F}^{g}_{A}]\,,\\ \widehat{\Pi}^{3\mathrm{p,g}}_{B} &= e_s\langle\bar{s}s\rangle\left(e^{-m_Q^2/M^2}-e^{-s_0/M^2}\right)\mathcal{J}_{\bar{u}_0}[\mathcal{F}^{g}_{B}]\,.\end{aligned} \tag{B6}$$

Appendix C: Explicit expressions for the invariant amplitudes $\widehat{\Pi}_{AA}$, $\widehat{\Pi}_{AB}$, and $\widehat{\Pi}_{BB}$

In this appendix, we give the OPE expressions used to determine the decay constants f_1 and f_2 of the axial-vector mesons from Eqs. (37) and (38).

$$\begin{aligned}
\widehat{\Pi}_{AA} = & \int_{(m_Q+m_s)^2}^{s_0} ds\, e^{-s/M^2} \rho_{AA}(s) \\
& + e^{-m_Q^2/M^2} \Bigg\{ \langle \bar{s}s \rangle \left[m_Q - \frac{m_s m_Q^2}{2M^2} + \frac{m_s^2 m_Q^3}{2M^4} \right] \\
& - \langle \bar{s} g_s \sigma \cdot G s \rangle \frac{m_Q^3}{4M^4} \Bigg\}, \qquad \text{(C1)} \\
\widehat{\Pi}_{AB} = & \int_{(m_Q+m_s)^2}^{s_0} ds\, e^{-s/M^2} \rho_{AB}(s) \\
& + e^{-m_Q^2/M^2} \Bigg\{ \frac{\langle \bar{s}s \rangle}{m_Q + m_s} \left[m_Q^2 + \frac{m_s m_Q}{2} \left(1 - \frac{m_Q^2}{M^2} \right) \right. \\
& \left. + \frac{m_s^2}{2} \left(\frac{m_Q^4}{M^4} - \frac{m_Q^2}{M^2} - 1 \right) \right] \\
& - \frac{\langle \bar{s} g_s \sigma \cdot G s \rangle}{4(m_Q + m_s)} \left(\frac{m_Q^4}{M^4} - \frac{m_Q^2}{M^2} - 1 \right) \Bigg\}, \qquad \text{(C2)} \\
\widehat{\Pi}_{BB} = & \int_{(m_Q+m_s)^2}^{s_0} ds\, e^{-s/M^2} \rho_{BB}(s) \\
& + e^{-m_Q^2/M^2} \Bigg\{ \frac{\langle \bar{s}s \rangle}{(m_Q + m_s)^2} \left[m_Q^3 + m_s m_Q^2 \left(1 - \frac{m_Q^2}{2M^2} \right) \right. \\
& \left. + m_s^2 \left(\frac{m_Q^5}{2M^4} - \frac{m_Q^3}{M^2} \right) \right] \\
& - \frac{\langle \bar{s} g_s \sigma \cdot G s \rangle}{(m_Q + m_s)^2} \left(\frac{m_Q^5}{4M^4} - \frac{m_Q^3}{2M^2} \right) \Bigg\}, \qquad \text{(C3)}
\end{aligned}$$

where

$$\rho_{AA}(s) = \frac{\sqrt{\lambda(s, m_Q^2, m_s^2)}}{8\pi^2} \left[2 - \frac{m_Q^2 + m_s^2 + 6 m_Q m_s}{s} - \frac{(m_Q^2 - m_s^2)^2}{s^2} \right], \qquad \text{(C4)}$$

$$\rho_{AB}(s) = \rho_{BA}(s) = \frac{\sqrt{\lambda(s, m_Q^2, m_s^2)}}{8\pi^2} \frac{3(m_s - m_Q)\left[(m_Q + m_s)^2 - s\right]}{(m_Q + m_s)s}, \qquad \text{(C5)}$$

$$\rho_{BB}(s) = -\frac{\sqrt{\lambda(s, m_Q^2, m_s^2)}}{8\pi^2} \frac{\left[(m_Q + m_s)^2 - s\right]\left(2m_s^2 - 4 m_Q m_s + 2m_Q^2 + s\right)}{(m_Q + m_s)^2 s}. \qquad \text{(C6)}$$

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